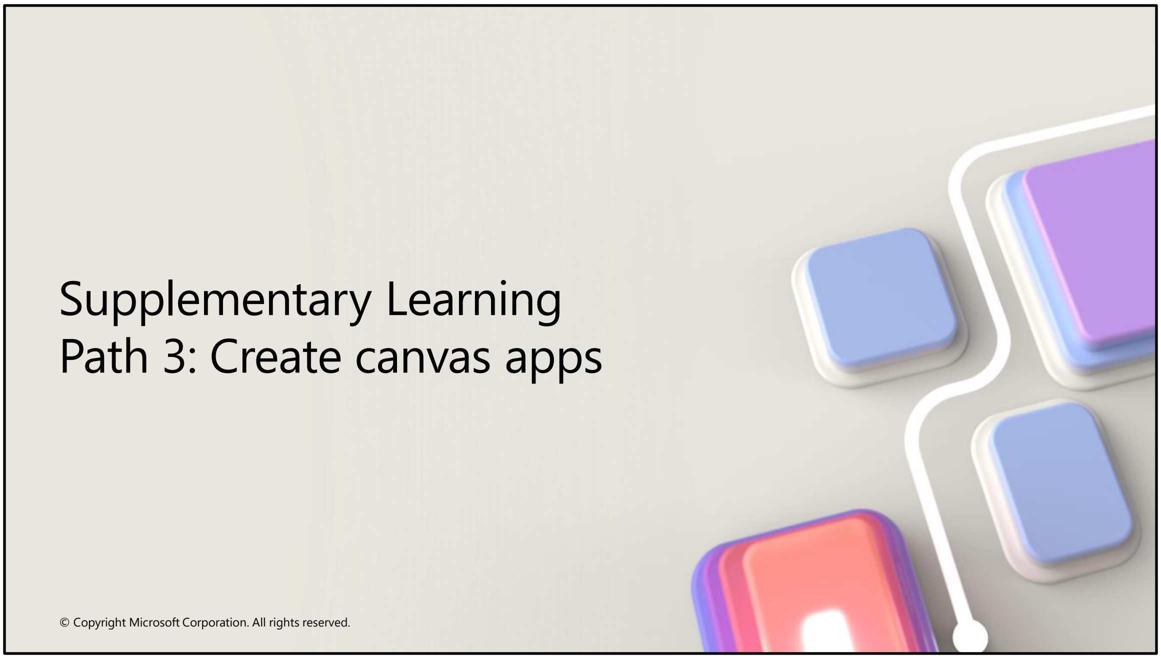
Microsoft

PL-400T00 Microsoft Power Platform Developer



This learning path covers creating canvas apps

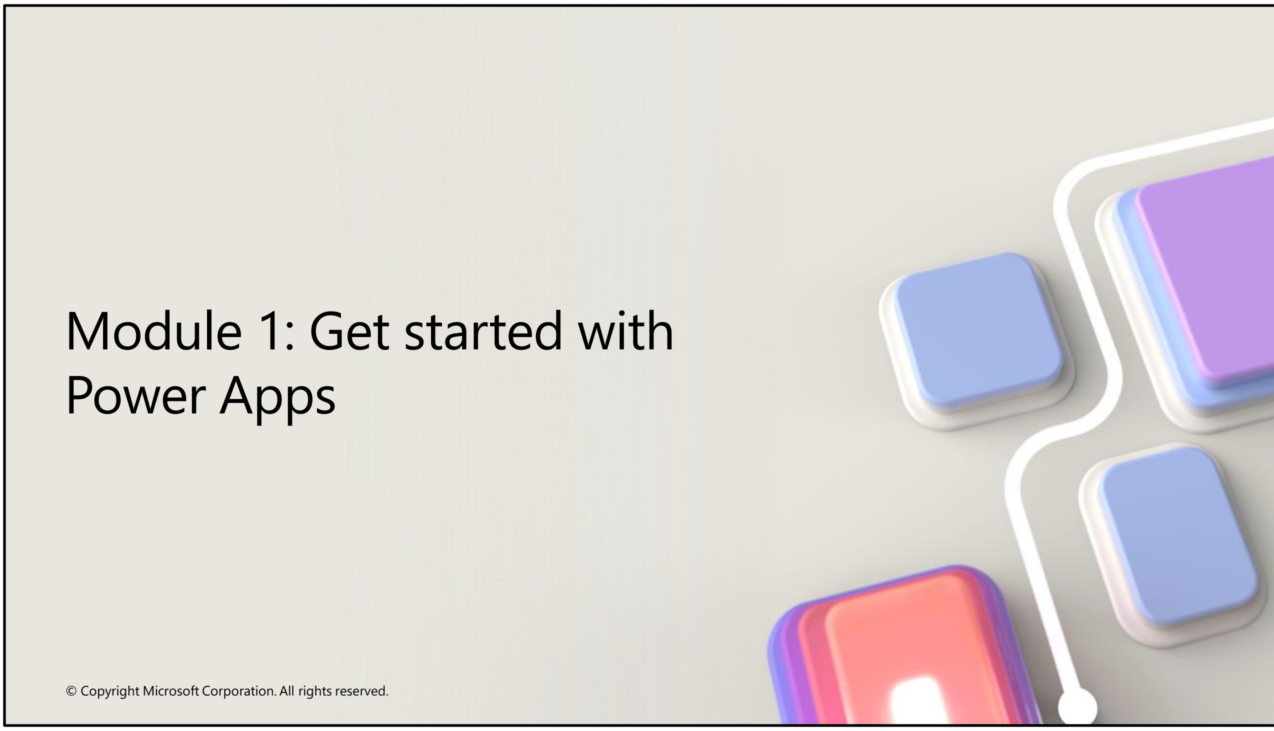
This learning path is optional as it is as these tasks are not the focus of the training/exam per the latest update to Objective Domain.

Modules

- Get started with Power Apps
- Understanding low code as a traditional developer
- Customize a canvas app
- Navigation in a canvas apps
- Power Fx formulas
- Canvas components
- Testing canvas apps

Modules

1. Get started with Power Apps
2. Understanding Low Code as a Traditional Developer
3. Customize a canvas app in Power Apps
4. Navigation in a canvas app in Power Apps
5. Power Fx formulas
6. Canvas components
7. Document and test your Power Apps application



We introduced Power Apps in the introduction module, and we will navigate ways to build Power Apps
Learn the basics of Power Apps

Learning Objectives

After completing this module, you will be able to:

1. Explore how Power Apps can make your business more efficient.
2. Learn how to use different technologies to perform different tasks in Power Apps.
3. Learn about the different ways to build an app in Power Apps.

Microsoft Learn module

Get started with Power Apps canvas apps

https://learn.microsoft.com/en-us/training/modules/get-started-with-powerapps/

Link to student courseware on Learn
Get started with Power Apps canvas apps
https://learn.microsoft.com/en-us/training/modules/get-started-with-powerapps/

Introduction to canvas apps

Power Apps is a suite of apps, services, connectors, and a data platform that provides you with an opportunity to build custom apps for your business needs.

- Build an app quickly by using the skills that you already have
- Connect to the cloud services and data sources that you’re already using
- Share your apps instantly so that coworkers can use them on their phones and tablets

Power Apps is a suite of apps, services, connectors, and a data platform that provides you with an opportunity to build custom apps for your business needs. By using Power Apps, you can quickly build custom business apps that connect to your business data that is stored either in the underlying data platform (Microsoft Dataverse) or in various online and on-premises data sources (SharePoint, Excel, Office 365, Dynamics 365, SQL Server, and so on).

Apps that are built by using Power Apps provide rich business logic and workflow capabilities to transform your manual business processes to digital, automated processes. Power Apps simplifies the custom business app building experience by enabling users to build feature-rich apps without writing code.

Power Apps also provides an extensible platform that lets pro developers programmatically interact with data and metadata, apply business logic, create custom connectors, and integrate with external data.

With Power Apps, you can:

Build an app quickly by using the skills that you already have.

Connect to the cloud services and data sources that you're already using.

Share your apps instantly so that coworkers can use them on their phones and tablets.

When it comes to using Power Apps to get things done and keep people informed, your options are nearly limitless. The following examples can help you think about how to use an app, instead of traditional paper notes, to run your business:

Equipment in the field - Often, company representatives who visit customers in the field carry clipboards to help guarantee a paper trail of parts with scheduled replacement dates. By running an app on a tablet, reps can look up the customer's equipment, see a picture of a part, test and analyze the part, and then order new parts. Reps can perform these tasks on-site instead of leaving the customer's warehouse.

Restaurant employee management - Employees of a large restaurant might fill out work schedules and vacation requests on a piece of paper that's affixed to a wall. With Power Apps running on everyone's smartphone, employees can open the app to record the same information, anywhere, anytime. The app can even send reminders for the start of the next day's shift.

If you're a beginner with Power Apps, this module gets you going quickly; if you're familiar with Power Apps, it ties concepts together and fills in the gaps.

Canvas apps building blocks

Tools and applications used

1. **Power Apps maker portal** – Apps start here, whether you build them from data, a sample app, or a blank screen
2. **Power Apps Studio** – Create and edit apps
3. **Power Apps mobile** – Run your apps on Microsoft Windows, Apple iOS, and Google Android devices

Power Apps is a collection of services, apps, and connectors that work together to let you do much more than just view your data. You can act on your data and update it anywhere and from any device.

This unit explores each part of the following Power Apps components:

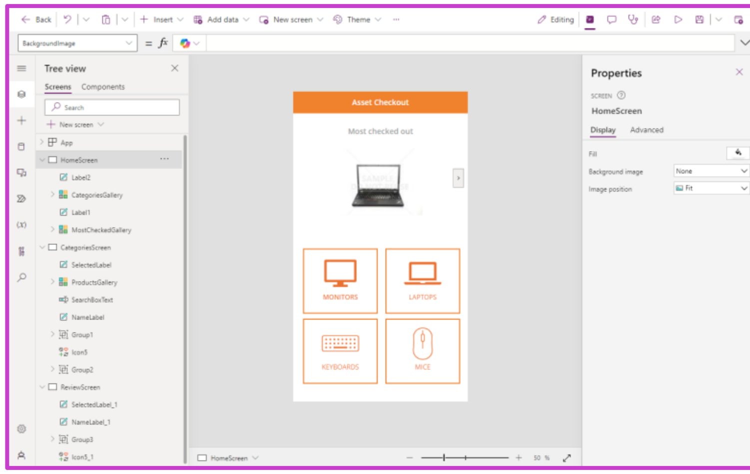
Power Apps Home Page - Apps start here, whether you build them from data, a sample app, or a blank screen.
Power Apps Mobile - Run your apps on Microsoft Windows, Apple iOS, and Google Android devices.
Power Platform Admin Center - Manage Power Apps environments and other components. – Already covered earlier in course

[!NOTE] To use these sites, you'll need to sign in by using your organizational account.

Power Apps Studio

Power Apps Studio is where you can fully develop your apps to make them more effective as a business tool and to make them more attractive.

- **Left pane** – Shows a hierarchical view of all the controls on each screen or a thumbnail for each screen in your app
- **Middle pane** – Shows the canvas app that you're working on
- **Right pane** – Where you set options such as the layout, properties, and data sources for certain controls



Power Apps Studio is where you can fully develop your apps to make them more effective as a business tool and to make them more attractive. Power Apps Studio has three panes that make creating apps seem more like building a slide deck in Microsoft PowerPoint:

Left pane - Shows a hierarchical view of all the controls on each screen or a thumbnail for each screen in your app.

Middle pane - Shows the canvas app that you're working on.

Right pane - Where you set options such as the layout, properties, and data sources for certain controls.

Power Apps can be a powerful tool for citizen developers and traditional developers alike. Upon finishing this module, a traditional developer will have learned how Power Apps work, what the formula language is, and how to create an app using Power Apps

In this module you will:

Understand what Power Fx is and how to use it

Create an app using Power Apps

Modify an app using Power Fx

Learning Objectives

After completing this module, you will be able to:

1. Understand what Power Fx is and how to use it
2. Create an app using Power Apps
3. Modify an app using Power Fx

Microsoft Learn module

Understanding low code as a traditional developer

https://learn.microsoft.com/training/modules/understanding-low-code-as-a-traditional-developer

Link to student courseware on Learn

Understanding Low Code as a Traditional Developer

https://learn.microsoft.com/training/modules/understanding-low-code-as-a-traditional-developer

What is low code?

Write small amount of code that would normally takes several more lines of code in a traditional programming environment

Power Fx formulas which have an Excel like syntax

Power Fx is a declarative language

Power Fx formula:

```
fx "Employee: " & LookUp( Orders, Number=Selected ).Employee.'First Name'
```

Employee: Elisa

JavaScript equivalent requires explicit: Asynchronous and lambda

```
RunQueryAsync( Orders,
    "$filter=nwind_ordernumber+eq+" + Escape(Selected) +
    "&$expand=nwind_EmployeeID($select=nwind_employeesid,nwind_firstname)" +
    "&$select=nwind_ordersid,nwind_ordernumber,nwind_EmployeeID,_nwind_employeeid_value" +
    "&$top=1"
).then(
    (result) => "Employee: " + result.nwind_firstname
);
```

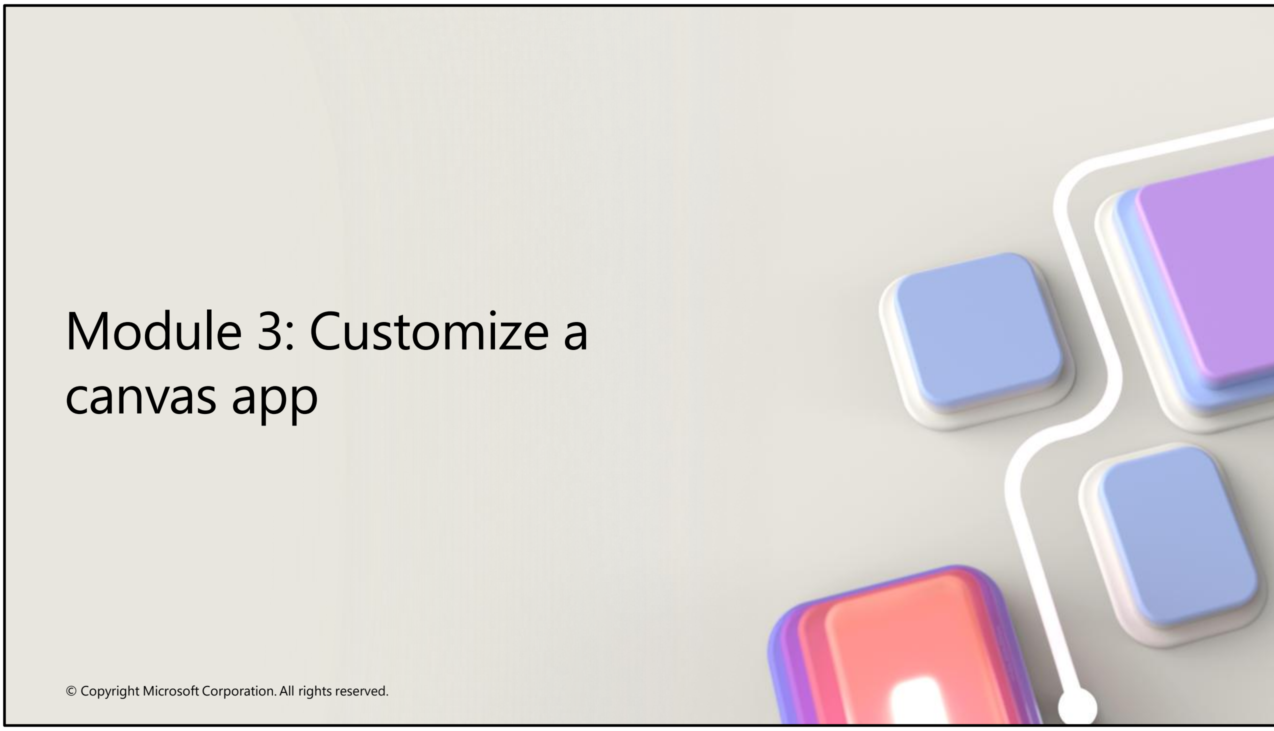

The term "low code" can mean different things to different people when they first hear it. What we mean when we talk about low code is that with tools like Power Apps, you only need to write a small amount of code to get results that would normally take several more lines of code in a traditional programming language.

In canvas-based Power Apps, the low code scripting language used is called Power Fx; that is the language we will be learning and using throughout this module to build our app.

Take this as an example: to look up the first name of the employee for an order, one would write the Power Fx formula or equivalent JavaScript as seen in the animation below. This animation shows the mapping between the parts of the Power Fx formula and the concepts that need to be explicitly coded in the equivalent JavaScript.

Many traditional application development stacks feature an *imperative* model for development across the stack. In other words, the app developer has to write code for everything that happens in the app: authentication & authorization, business logic, the user interface, server or cloud service configuration, and so on.

Power Fx is a declarative language, just as Excel is. The developer defines what behavior they want, but it's up to the system to decide how and when to accomplish it. To make that practical, most work is done through pure functions without side effects, making Power Fx also a functional language.



Learn how to customize an app by adding controls.

In this module, you will:
- Create an app
- Explore controls
- Explore Galleries

Learning Objectives

After completing this module, you will be able to:

1. Change the layout of a gallery.
2. Change the control with which a user provides information.
3. Explore controls on each screen of an app.

Microsoft Learn module

Customize a canvas app in Power Apps

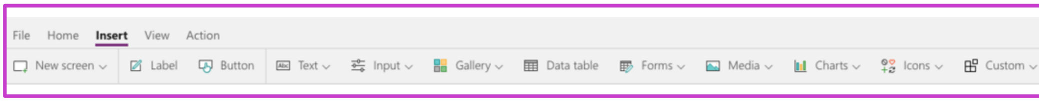
https://learn.microsoft.com/training/modules/customize-apps-in-powerapps

Link to student courseware on Learn

Customize a canvas app in Power Apps

https://learn.microsoft.com/training/modules/customize-apps-in-powerapps

Controls in Power Apps



UI elements that produces an action or shows information

Common controls include:

Galleries:
Layout containers that hold a set of controls that show rows from a data source

Forms:
Show details about your data and let you create and edit rows

Label:
A label shows data that you specify as a literal string of text, which appears exactly the way you type it, or as a formula that evaluates to a string of text

Text Input:
A box in which the user can type text, numbers, and other data

A control is a UI element that produces an action or shows information. Many controls in Power Apps are similar to controls that you've used in other apps: labels, text-input boxes, drop-down lists, navigation elements, and so on.

In addition to these typical controls, Power Apps has more specialized controls, which you can find on the Insert tab.

A few controls that can add interest and impact to your apps include:

Galleries - These controls are layout containers that hold a set of controls that show rows from a data source.

Forms - These controls show details about your data and let you create and edit rows.

Media - These controls let you add background images, include a camera button (so that users can take pictures from the app), a barcode reader for quickly capturing identification information, and more.

Charts - These controls let you add charts so that users can perform instant analysis while they're on the road.

To see what controls are available, select the Insert tab, and then select each option in turn.

Demo before exercise

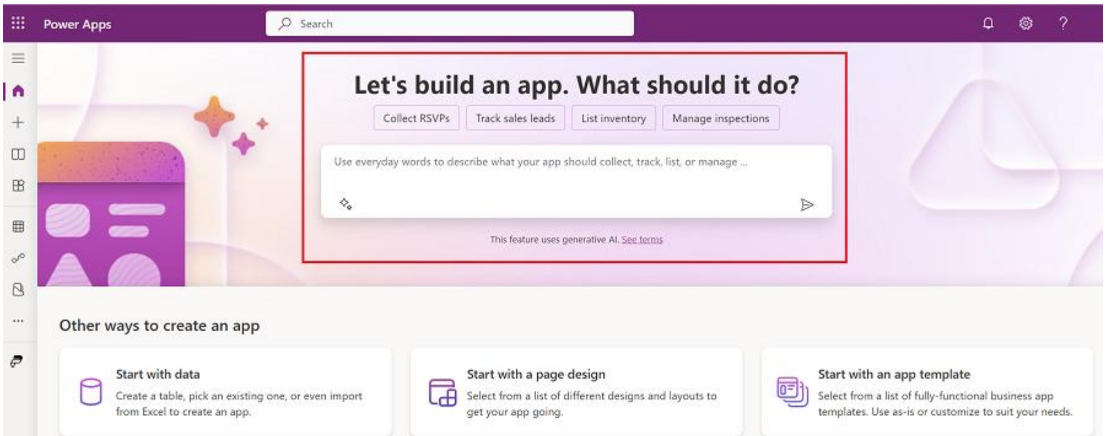
Create and customize a canvas app in Power Apps

DEMO

Demo

1. Create a blank canvas app
2. Rename Screen1 to MainScreen
3. Add a header
    1. Add rectangle
        1. Rename
        2. Set Fill Color
    2. Add label
        1. Rename
        2. Set Text to Title of app
        3. Set Font color to White
        4. Move to top right of screen
    3. Resize rectangle
        1. Move to X=0, Y=0
        2. Set Width=Parent.Width
        3. Set Height = Parent.Height /10
    4. Group rectangle and label
4. Add a Dataverse table to the app
5. Create a Gallery for the table
    1. Rename
    2. Change layout
    3. Change fields
    4. Set X, Y, Width and Height relative to Header

Create an app from conversations



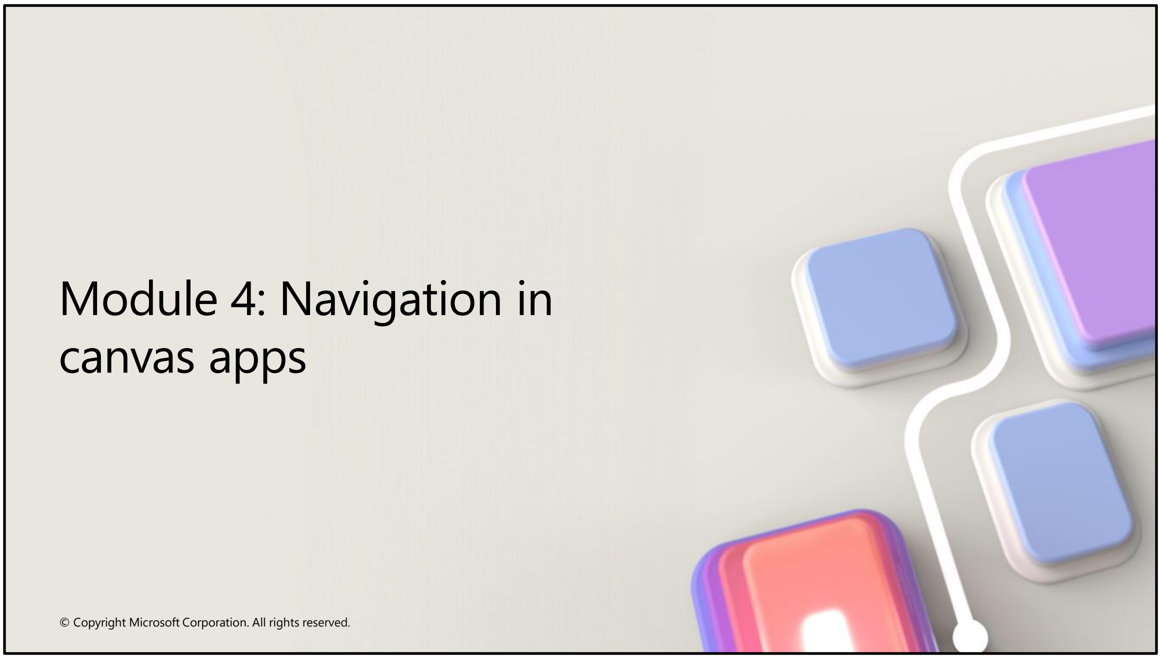
Create Power Apps with the help of AI. Describe the app that you want to build, and AI will design it for you.

With the **Copilot** feature in Power Apps, you get in-app guidance using natural language processing to help you build your app.

The AI assistant is available from the Power Apps home screen. You can tell the AI assistant what kind of information you want to collect, track, or show and the assistant will generate one or more Dataverse tables along with relationships and use it to build your canvas app.

This works best as an additional instructor demo. It's also recommended demonstrate the new capabilities for reviewing / modifying more complex table schemas (currently in preview as of June 2024; good opportunity to mention the https://make.preview.powerapps.com/ URL for experimenting with new functionality)

https://learn.microsoft.com/en-us/power-apps/maker/canvas-apps/ai-conversations-create-app#build-apps-through-conversation-general-availabilible-ga-version



App users can only navigate through the navigation options provided by an app developer, and this module is designed to help you build a good navigation experience for your canvas app.

In this module, you will:

- Understand how navigation works in a canvas app
- Use the Navigate and Back functions

Learning Objectives

After completing this module, you will be able to:

1. Understand how navigation works in a canvas app
2. Use the Navigate and Back functions
3. Understand the different ways these functions can be invoked

Microsoft Learn module

Navigation in a canvas app in Power Apps

https://learn.microsoft.com/training/modules/navigation-canvas-app

Link to student courseware on Learn

Navigation in a canvas app in Power Apps

https://learn.microsoft.com/training/modules/navigation-canvas-app

Understanding navigation

Since most apps have multiple screens, understanding how to implement the Navigate function in your app is critical

Typically added to the OnSelect property of a Button or Icon control

Specifies which screen transition effect if any, should be used in the app to navigate to

Navigate(Screen2, ScreenTransition.Cover)

Specifies which screen in the app to navigate to

In Power Apps, because most apps have multiple screens, it's essential that you understand how to implement the Navigate function in your app. The Navigate function allows users to move or navigate from screen to screen in an app. For example, if you create an app with three screens and you want to give users a way to navigate to another screen, set the OnSelect property of a Button control to:

Navigate(Screen2,ScreenTransition.Cover)

With this formula, when the button is selected, Screen2 will automatically display. You could also use this formula on an icon, like an arrow, to provide navigation in the app.

The Navigate function also allows for a visual transition as the users move from screen to screen, which is set using ScreenTransition. In the example shown above, ScreenTransition.Cover was applied. There are a few different ScreenTransitions to choose from, and each one provides a slightly different user experience when navigating screens. ScreenTransitions will be covered in further detail later in this module. Similar to the Navigate function, you also have the option to use the Back() function. The Back() function is fairly straight-forward, it sends the user back to the previous screen.

You can use Navigate to set one or more context variables. You can use this approach to pass parameters to a screen. If you've used another programming tool, this is similar to passing parameters to procedures.

Demo: Use Back and Navigate functions

DEMO

1. Add Blank Screen
    1. Rename to DetailScreen
    2. Copy and Paste Header from MainScreen to DetailScreen
2. In Gallery
    1. OnSelect Navigate(DetailScreen)
3. Add Left icon to Detail Screen
    1. On Select = Back()

Power Fx formulas are used to

- Set Values
- Format output
- Perform calculations
- Control visibility and read/edit mode
- Set Position of controls
- Filter and search data from data sources

Learning Objectives

After completing this module, you will be able to:

1. Create formulas to change properties in a Power Apps canvas app
2. Create formulas to change behaviors in a Power Apps canvas app
3. Author a basic formula that uses tables and records in a Power Apps canvas app

Microsoft Learn module

Create formulas to change properties in a Power Apps canvas app
https://learn.microsoft.com/training/modules/author-basic-formula-change-properties-powerapps

Create formulas to change behaviors in a Power Apps canvas app
https://learn.microsoft.com/training/modules/author-basic-formula-change-behaviors-powerapps

Author a basic formula that uses tables and records in a Power Apps canvas app
https://learn.microsoft.com/training/modules/author-basic-formula-tables-records-powerapps

Link to student courseware on Learn

Create formulas to change properties in a Power Apps canvas app
https://learn.microsoft.com/training/modules/author-basic-formula-change-properties-powerapps
Create formulas to change behaviors in a Power Apps canvas app
https://learn.microsoft.com/training/modules/author-basic-formula-change-behaviors-powerapps
Author a basic formula that uses tables and records in a Power Apps canvas app
https://learn.microsoft.com/training/modules/author-basic-formula-tables-records-powerapps

Power Fx formulas

```
fx NewForm(EditForm1);Navigate(EditScreen1, None)
```

When using Microsoft Power Apps, you don’t have to write complicated application code the way that a traditional developer does.

Configure your app with formulas that not only calculate values and perform other tasks (as they do in Excel) but also respond to user input.

For example, you build a formula to determine how your app responds when users select a button, adjust a slider, or provide other input.

https://docs.microsoft.com/en-us/powerapps/maker/canvas-apps/formula-reference

Power Fx

What can formulas be used for in canvas apps

1. Set values
2. Format output
3. Perform calculations
4. Control visibility and display mode of controls
5. Set the position of controls
6. Filter data
7. And much much more

- Set Values
- Format output
- Perform calculations
- Control visibility and read/edit mode
- Set Position of controls
- Filter and search data from data sources

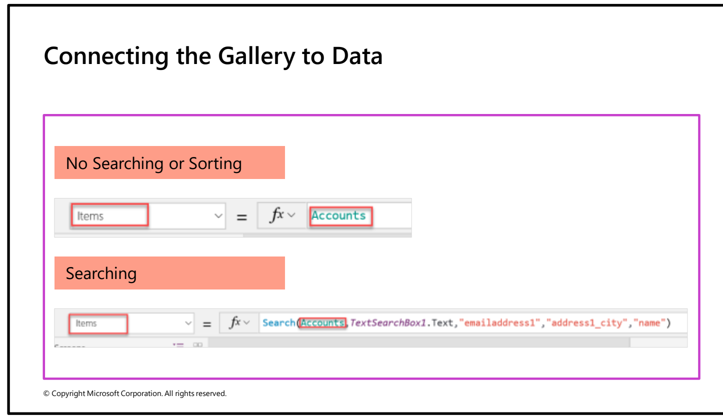
Core properties of controls

While each control may have slightly different properties the following are important ones to be aware of:

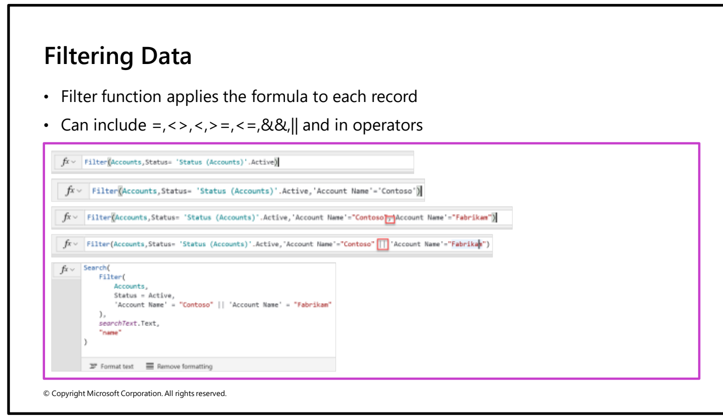
1. **Default**: Initial value of control before it is changed by the user
2. **Items**: The source of data that appears in a control such as a gallery, list, or chart
3. **OnChange**: How the app responds when the user changes the value of a control. For example, when a user selects a different value in a Dropdown control
4. **OnSelect**: How the app responds when the user taps or clicks a control
5. **Text**: Text that appears on a control or that the user types into a control
6. **Tooltip**: Explanatory text that appears when the user hovers over a control
7. **Visible**: Whether a control appears or is hidden

Because controls are designed with specific use cases in mind, the properties for each control are slightly different. Here are some important properties to be aware of:

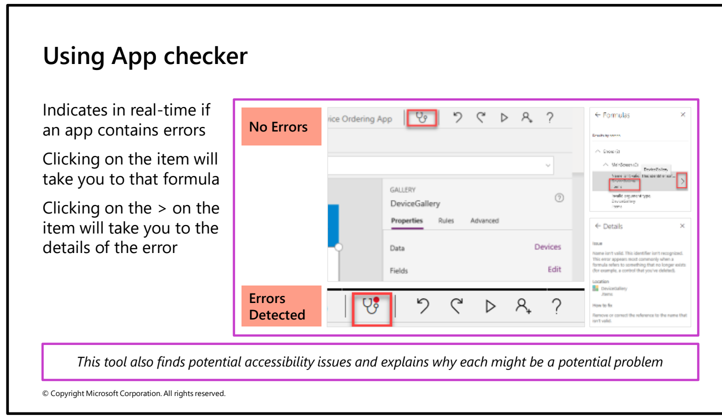
- Default - The initial value of a control before it is changed by the user. For example, when working with a Drop down control you could set the default value to appear when users see the control.
- DelayOutput - Set to true to delay action during text input.
- DisplayMode - Values can be Edit, View, or Disabled. Configures whether the control allows user input (Edit), only displays data (View), or is disabled (Disabled). For more information about display modes, see Use basic formulas to make better canvas apps in Power Apps learning path.
- Items - The source of data that appears in a control such as a gallery, list, or chart.
- OnChange - How the app responds when the user changes the value of a control. For example, when a user selects a different value in a Dropdown control.
- OnSelect - How the app responds when the user taps or clicks a control.
- Reset - Whether a control reverts to its default value. For more information, see Reset function in Power Apps.
- Text - Text that appears on a control or that the user types into a control.
- Tooltip - Explanatory text that appears when the user hovers over a control.
- Visible - Whether a control appears or is hidden.



The search in the related list will work on emailaddress1, address1_city and name while the sorting will be done on emaiaddress1.



3rd one with "," highlight how that creates an illogical condition that can't happen but it will still let you build it

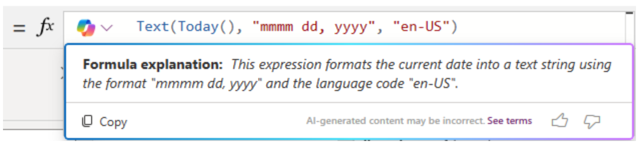


We added these extra to expand for exam coverage

App Checker in real time indicates there are errors
Clicking on the item will take you to that formula
Clicking on the > on the item will take you to the details of the error

Explain or create a formula using Copilot

• Use Copilot to explain what a formula is doing or create a new one, based on a comment

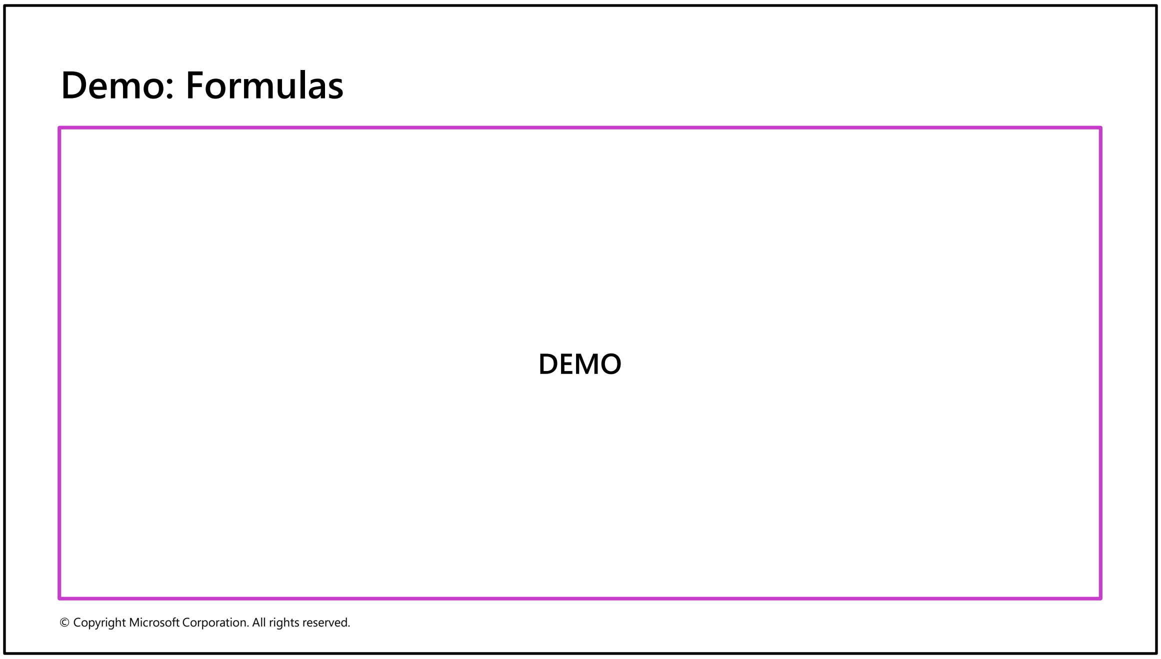


Use Copilot in Power Apps to create and modify Power Fx formulas quickly. You can use Copilot in the formula bar to help explain Power Fx formulas in natural language, or create Power Fx formulas from natural language.

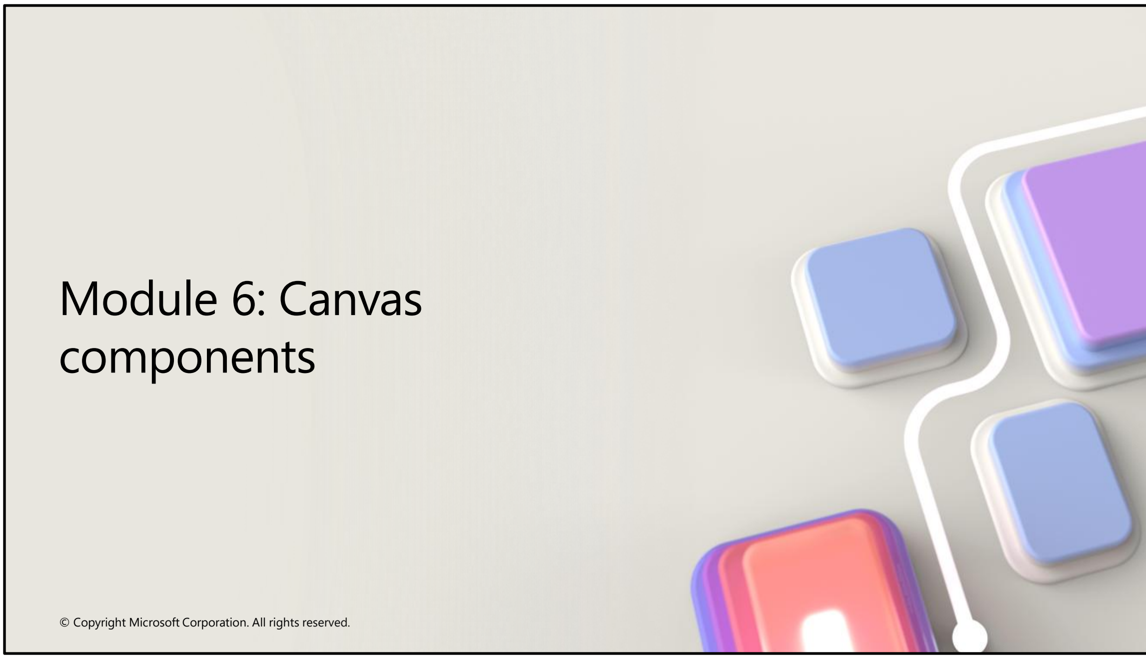
https://learn.microsoft.com/en-us/power-apps/maker/canvas-apps/ai-formulas-formulabar

Known Limitations

- Code comments only work with general Power Fx functions, and not Power Apps specific functions such as Navigate().
- While Copilot is generating formula suggestions, other activities in the Power Apps Studio might be temporarily unavailable.
- User defined functions aren't supported.
- Availability may be restricted based on region; consult the Learn documentation for further details



1. In OnStart property
    1. Set (userprofile, User()); Clear(selectedRecords)
    2. Run OnStart
2. Add label
    1. Set Text to userprofile.FullName
3. Items property for the gallery
    1. Filter the rows from the Dataverse table
4. OnSelect property for the gallery
    1. Collect(selectedRecords, ThisItem)
5. Add Gallery to DetailsScreen
    1. Items = selectedRecords
6. Preview the app
    1. Select a row from the gallery
    2. Back and select another row
    3. Show Collection



In this module will explain

- Canvas components
- Component libraries

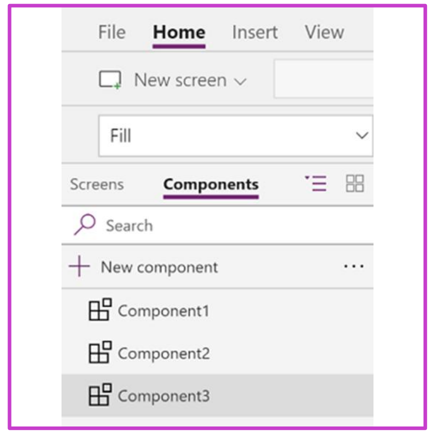
Learning Objectives

After completing this module, you will be able to:

1. Work with canvas components
2. Create component libraries

Canvas Components

- Reusable building blocks
- Can be used multiple times
- Custom properties define interface with hosting screen
- Component libraries



Components are reusable building blocks for canvas apps so that app makers can create custom controls to use inside an app, or across app using component library. Components can use advanced features, such as custom properties and enable complex capabilities.

Components are useful in building larger apps that have similar control patterns. If you update a component definition inside the app, all instances in the app reflect your changes. Components also reduce duplication of efforts by eliminating the need to copy/paste controls and improve performance.

Components also help create collaborative development and standardizes look-and-feel in an organization when you use a component library.

https://learn.microsoft.com/en-us/power-apps/maker/canvas-apps/create-component
https://learn.microsoft.com/en-us/power-apps/maker/canvas-apps/component-library

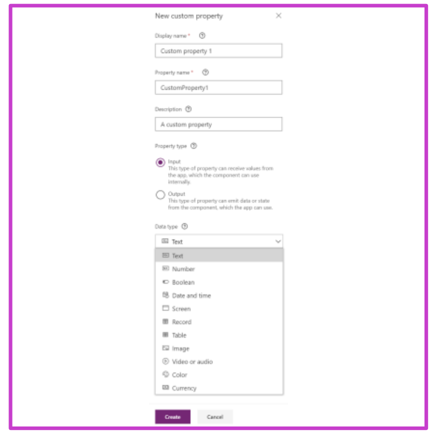
Component custom properties

Can be Input or Output

Default value is assigned based on data type

Enable Access app scope to share with component:

- Global variables
- Collections
- Controls on screens
- Tabular data sources



A component can receive input values and emit data if you create one or more custom properties.

Input property is how a component receives data to be used in the component. Input properties appear in the **Properties** tab of the right-hand pane if an instance of the component is selected. You can configure input properties with expressions or formulas, just as you configure standard properties in other controls. Other controls have input properties, such as the **Default** property of a **Text input** control.

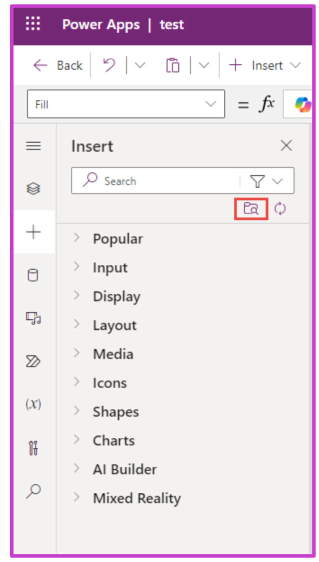
Output property is used to emit data or component state. For example, the **Selected** property on a **Gallery** control is an output property. When you create an output property, you can determine what other controls can refer to the component state.

When **Access app scope** is turned on, the following are accessible from within a component:

- Global variables
- Collections
- Controls and components on screens, such as a TextInput control
- Tabular data sources, such as Dataverse tables

Component Libraries

- Repository of components for reusability
- Discover and search components
- Publish updates
- Notify app makers of available component updates
- Import from a component library
- Notification in Power Apps Studio when an updated component in the library is available



A component library provides a centralized and managed repository of components for reusability.

Component libraries are containers of component definitions that make it easy to:
- Discover and search components
- Publish updates
- Notify app makers of available component updates

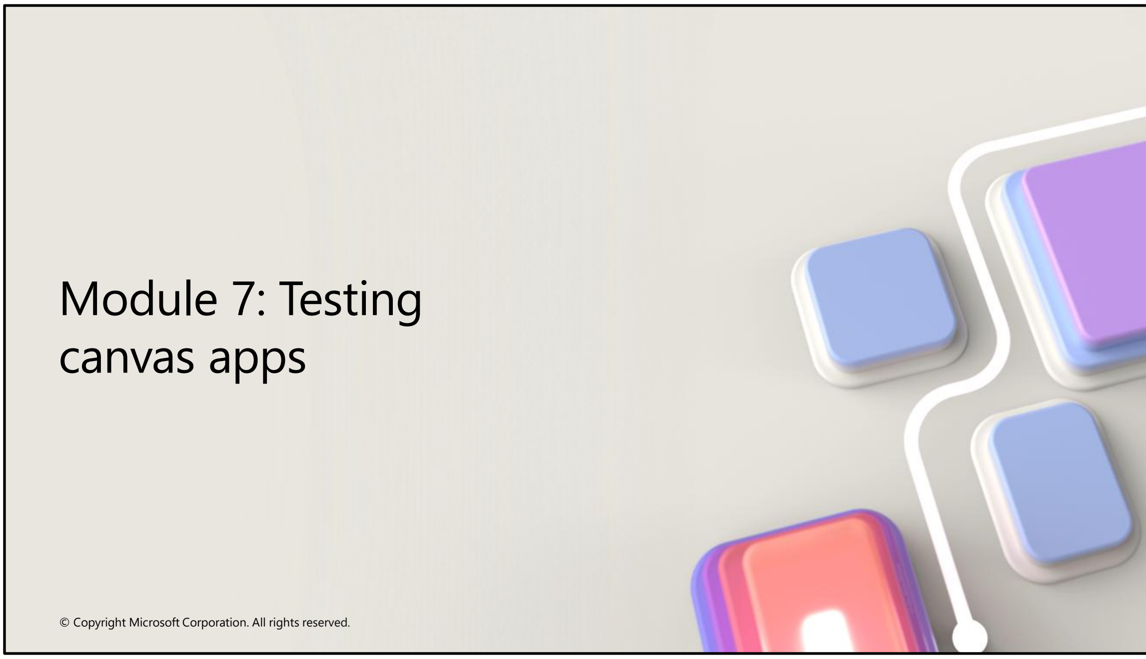
https://learn.microsoft.com/en-us/power-apps/maker/canvas-apps/component-library

Note: Components in a component library can never have access to app scope

Demo: Create components

DEMO

https://learn.microsoft.com/en-us/power-apps/maker/canvas-apps/create-component#create-an-example-component



In this module you will learn about Test Studio

Learning Objectives

After completing this module, you will be able to:

1. Learn about the different types of test plans and components of a good test plan
2. Identify and discuss optimization tools and performance tuning
3. Learn about the benefits of documenting your application

Microsoft Learn module

Document and test your Power Apps application

https://learn.microsoft.com/training/modules/document-test-powerapps-app

Link to student courseware on Learn

Document and test your Power Apps application

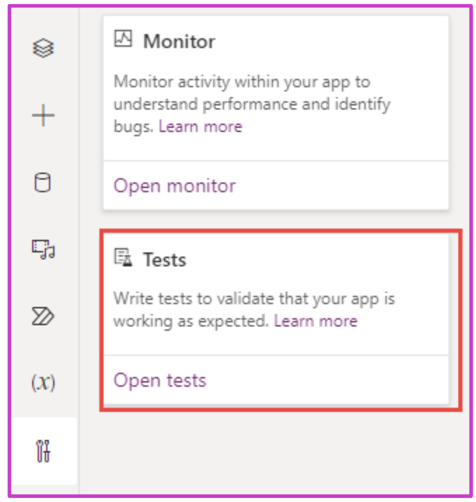
https://learn.microsoft.com/training/modules/document-test-powerapps-app

Power Apps Test Studio

Automate the testing of canvas apps

Low code test tool

- Record app interactions
- Write expressions
- Run tests
- Add tests to ALM process



Power Apps Test Studio is a low-code solution to write, organize, and automate tests for canvas apps. In Test Studio, you can write tests using Power Apps expressions or use a recorder to save app interaction to automatically generate the expressions. You can play written tests back within the Test Studio to validate app functionality, and also run the tests in a web browser and build the automated tests into your app deployment process.

https://learn.microsoft.com/en-us/power-apps/maker/canvas-apps/test-studio

Power Apps Test Studio

Components of Test Studio

1. Test Suites
2. Test Cases
3. Test Steps
4. Test Assertions

Test cases

Test cases are made up of a series of instructions or actions, called test steps. Test cases are executed to validate that your app, or specific features in your app, is working as you expect. For example, in an Expense app, you would like to ensure that only expenses with associated actual cost can be submitted. A test case can help verify that this condition or requirement is always met.

In Test Studio, test steps are written using the Power Apps expression language. Test expressions can consist of both the functions available when building your app and additional expressions to support automated testing.

Test suites

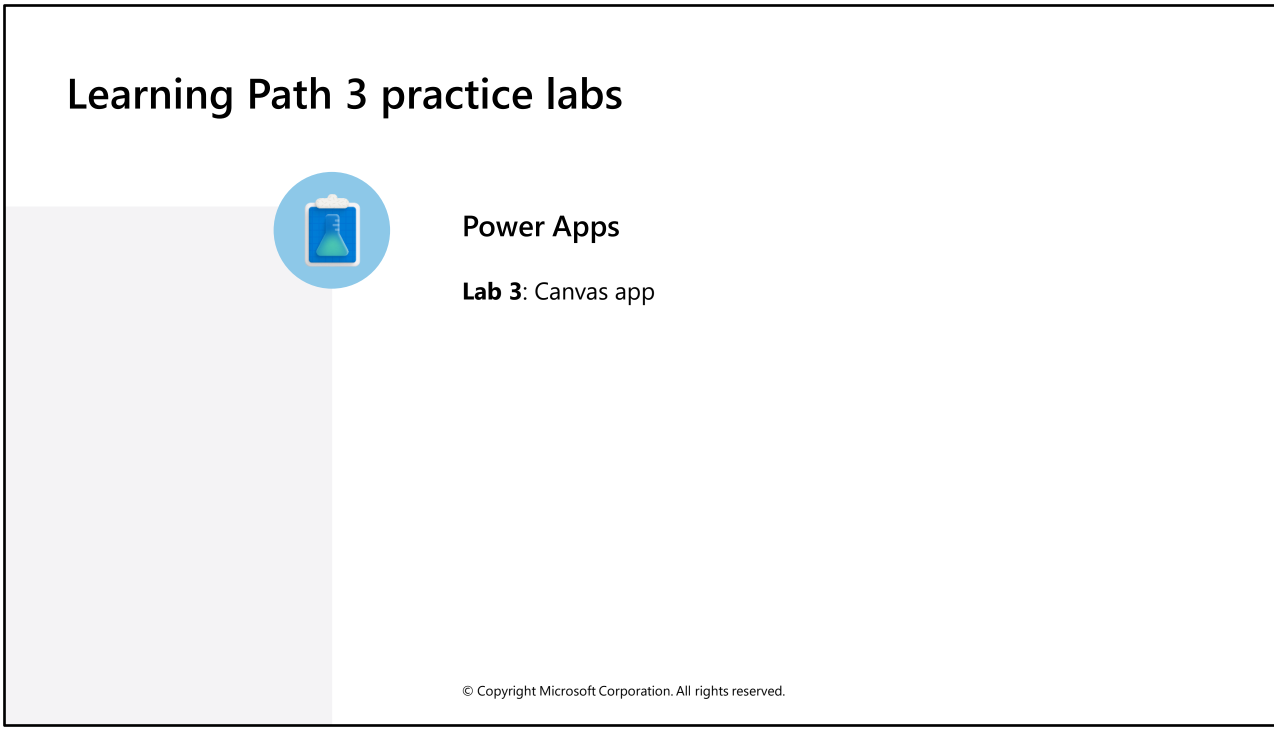
Test suites are used to organize or group test cases together. As the number of test cases in the app grows, you might consider organizing the test cases in specific features or functionality. For example, you might have one test suite with test cases to validate expense report submissions and another test suite that focuses just on expense approvals.

Test cases contained in test suites are run sequentially. The app state is persisted across all test cases in a suite. For example, if you have a test case that completes on Screen 5 in your app, the next test case in your test suite will begin running from Screen 5. It allows you to break down a complex test scenario into multiple test cases within a single suite, and the state is shared across all test cases. If your second test case expects to begin at the start screen of the app, you can navigate to the start screen as the first step in your test case. It's important to remember that the app is not reloaded at the beginning of every test case in a test suite when planning your test execution.

Test assertions

Every test case should have an expected result. To validate the expected result of a test against the actual result of your test, you can write test assertions. An assertion is an expression that evaluates to true or false in the test. If the expression returns false, the test case will fail.

https://learn.microsoft.com/en-us/power-apps/maker/canvas-apps/test-studio



As we continue to build our solution, we will now design a Power Apps canvas app that the inspectors will use in the field on their mobile devices. Canvas apps are low code apps that can be designed for a tablet or mobile phone layouts. You will build a two-screen canvas app that allows inspectors quickly access and process the inspections.

Summary

- Canvas apps give you pixel precision to build smart apps using low code, or no code
- You can create an app by using several different methods. Some of these methods include from a template, a data source (like Microsoft SharePoint), or a blank canvas
- Canvas apps are available across devices and platforms
- You build canvas apps by adding controls to screens and setting the properties of the controls
- By modifying various control properties, you can completely change how the control looks and functions when users interact with the app
- Power Fx formulas are used on the properties of controls
- Create Component libraries for reuse
- Test Studio performs regression testing

Modules

1. Get started with Power Apps
2. Understanding Low Code as a Traditional Developer
3. Customize a canvas app in Power Apps
4. Navigation in a canvas app in Power Apps
5. Power Fx formulas
6. Canvas components
7. Test your Canvas app

Microsoft